Synthetic Bernoulli Data for Task II

Fixed image embeddings and nonlinear annotator bias

The simulation retains the empirical distribution of real image embeddings but replaces the disease labels with 27 synthetic binary aspects. Each aspect has a shared latent probability μ_h^* and annotator-specific probabilities $\mu_{r,h}$, related through the logit-space bias $b_{r,h}$. The notation follows the attached Bernoulli framework [1], with h indexing aspects. The current design uses 34 annotators, $\tau = 0.1$, and an annotator-population positive-rate target of 0.5.

1 Input distribution and shared latent tasks

Retain all distinct images in the Task II pool with matched pooled embeddings, denoted by $x_1, \ldots, x_M \in \mathbb{R}^{1376}$. Their uniform empirical distribution defines P_X, so every item expectation in the construction is a full-pool average:

$$\mathbb{E}_X q(X) = \frac{1}{M} \sum_{i=1}^{M} q(x_i).$$

The real disease labels are not used. Apply PCA to the full embedding pool, retain the first ten components, and standardize them to obtain a fixed feature map $z(x) \in \mathbb{R}^{10}$ with $\mathbb{E}_X z(X) \approx 0$ and $\mathrm{Cov}_X\, z(X) \approx I_{10}$.

For each aspect $h = 1, \ldots, 27$, independently draw $c_h \sim N(0, I_{10})$ and normalize it to unit Euclidean norm. Define

$$s_h(x) = c_h^\top z(x), \qquad \mu_h^*(x) = \sigma\{a_h + s_h(x)\}, \qquad \sigma(t) = \frac{1}{1 + e^{-t}}. \tag{1}$$

The intercept a_h controls prevalence and is specified below. These are synthetic, nonexclusive binary tasks, not the original disease categories. The same $z(x)$ is used for both the shared signal and the annotator-specific nonlinear bias.

2 Annotator effects, probabilities, and labels

For each annotator–aspect pair (r, h), independently draw a random network $f_{r,h}$ with architecture $10 \to 32 \to 1$, a tanh hidden layer, and a linear output. Independently draw a global offset $C_{r,h} \sim N(0, \tau^2)$. Each network and offset is drawn once and then held fixed across all images. Set

$$b_{r,h}(x) = C_{r,h} + f_{r,h}(x) - \mathbb{E}_X f_{r,h}(X). \tag{2}$$

Centering uses the entire fixed image pool, hence $\mathbb{E}_X b_{r,h}(X) = C_{r,h}$ exactly. Thus $C_{r,h}$ describes the annotator's overall signed logit offset, whereas the centered network changes its bias across images without changing that overall offset. The network moment conditions needed for the claimed bias bound are stated in Section 3.

The annotator's conditional probability and observed label are

$$\begin{aligned} \mu_{r,h}(x) &= \sigma\{a_h + s_h(x) + b_{r,h}(x)\}, \\ Y_{r,h}(x) \mid x, \xi_r &\sim \mathrm{Bernoulli}\{\mu_{r,h}(x)\}, \end{aligned} \tag{3}$$

where ξ_r collects annotator r's random effects. Consequently, $\mathrm{logit}\, \mu_{r,h}(x) = \mathrm{logit}\, \mu_h^*(x) + b_{r,h}(x)$, exactly as in the attached model. Conditional on the inputs and all annotator effects, labels are generated independently across annotators and, by the stated simulation rule, across aspects.

After specifying the bias-generating distribution, choose a_h by solving

$$\mathbb{E}_X \mathbb{E}_r\, \sigma\{a_h + s_h(X) + b_{r,h}(X)\} = 0.5. \tag{4}$$

Here $\mathbb{E}_r$ averages over the annotator population, not over the 34 realized annotators. The left-hand side is continuous and strictly increasing in a_h, with limits zero and one, so the population equation has a unique solution. It targets the average annotator probability; neither $\mathbb{E}_X \mu_h^*(X)$ nor the realized fraction of positive labels is required to equal 0.5. The supplied workflow does not specify the numerical approximation used for $\mathbb{E}_r$ in this solve.

3 Bias moments and the bound $\kappa = 2$

The theoretical requirements are population centering $\mathbb{E}_r b_{r,h}(x) = 0$ for every fixed x, Gaussian global offsets, and the sensitivity condition

$$\mathbb{E}_r \operatorname{Var}_X\{b_{r,h}(X)\} \le (\kappa - 1)\mathbb{E}_r\{\mathbb{E}_X b_{r,h}(X)\}^2 = (\kappa - 1)\tau^2. \tag{5}$$

Centering over X in (2) identifies $C_{r,h}$ but does not, by itself, establish population centering. The random-network distribution must also be centered.

Required output-weight moments. Write the random network, omitting an output intercept that cancels in (2), as

$$f_{r,h}(x) = \sum_{j=1}^{32} v_{r,h,j}\phi_{r,h,j}(x), \qquad |\phi_{r,h,j}(x)| \le 1,$$

where $\phi_{r,h,j}$ is the output of hidden tanh unit j and $v_{r,h,j}$ is its linear output weight. For the claimed $\kappa = 2$ bound, interpret the stated output-layer scale $\tau^2/32$ as a *variance bound*: conditional on the hidden layer, the output weights must be independent, have mean zero, and have variance at most $\tau^2/32$. These are explicit sufficient conditions; the supplied workflow does not fully specify the weight-initialization distribution.

Under these conditions, $\mathbb{E}_r f_{r,h}(x) = 0$. Together with the independently drawn centered offset and full-pool centering, this gives

$$\mathbb{E}_r b_{r,h}(x) = 0 \quad \text{for every fixed } x, \qquad C_{r,h} = \mathbb{E}_X b_{r,h}(X) \overset{\text{i.i.d.}}{\sim} N(0, \tau^2) \quad \text{across } r.$$

The centering is on the logit scale; it does not imply $\mathbb{E}_r \mu_{r,h}(x) = \mu_h^*(x)$ after the nonlinear sigmoid transformation.

Derivation of the sensitivity bound. The offset and the centering constant do not affect variance over X. Conditional independence and zero means of the output weights remove the cross terms, while bounded hidden outputs imply $\operatorname{Var}_X\{\phi_{r,h,j}(X)\} \le 1$. Therefore,

$$\begin{aligned}\mathbb{E}_r \operatorname{Var}_X\{b_{r,h}(X)\} &= \mathbb{E}_r \operatorname{Var}_X\{f_{r,h}(X)\} \\ &\le \frac{\tau^2}{32}\sum_{j=1}^{32} \mathbb{E}_r \operatorname{Var}_X\{\phi_{r,h,j}(X)\} \le \tau^2.\end{aligned}$$

Since $\mathbb{E}_r C_{r,h}^2 = \tau^2$, condition (5) holds with the common choice $\kappa = 2$ for every aspect, and

$$\mathbb{E}_X \mathbb{E}_r b_{r,h}(X)^2 = \mathbb{E}_r \operatorname{Var}_X\{b_{r,h}(X)\} + \mathbb{E}_r C_{r,h}^2 \le 2\tau^2.$$

Thus $\tau = 0.1$ is a generating parameter, whereas $\kappa = 2$ is an analytic population upper bound implied by the stated moment conditions, not an additional input for generating labels. A variance ratio computed from only the 34 realized annotators is an empirical diagnostic, not this population bound.

Generated data and connection to Task II. The resulting observations are $Y_{r,h}(x_i)$ for $i = 1, \dots, M$, $r = 1, \dots, 34$, and $h = 1, \dots, 27$; the simulator also knows μ_h^*, $\mu_{r,h}$, and $b_{r,h}$. Two annotator identities serve as A, B, leaving a full reference set R of size $K = 32$ with $R \cap \{A, B\} = \varnothing$. Subsequently, each annotator's labels train a 27-output surrogate $\widehat{\mu}_r$ using unweighted Bernoulli cross-entropy, targeting μ_r rather than μ^*. Out-of-sample predictions then enter the reference-based analysis in [1].

Source. Construction and numerical settings: the supplied agent workflow. [1] *Task 2 under Bernoulli Labels and Logit-Space Annotation Bias: Complete corrected framework and derivation*, attached note, Section 1, equations (1)–(2).